

BIO-302 OBJECTIVES

FINAL TERM- 100%

GUESS PAPER 2023

MOLECULAR BIOLOGY

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OBJECTIVES

- ❖ RNA is single strand when just formed DNA _____ while **DNA is double-stranded.**
- ❖ Some proteins contain two or more separate polypeptide chains or subunits, which may be identical or different. _____ **Quaternary Structure of Proteins**
- ❖ DNA polymerase-pol II to..... **Metal ions Mg^{2+} OR Zn^{2+}**
- ❖ separation of DNA is accomplished by _____ **DNA helicases**
- ❖ After **cisplatin enters** the cell by passive diffusion or active transport, it undergoes hydrolysis to produce a highly reactive and charged complex (**$Pt(NH_3)_2ClH_2O^+$**).
- ❖ RNA polymerase can initiate a new RNA chain on a DNA template and thus does not **need a primer.**
- ❖ Rho-independent terminators as also called **intrinsic terminators**
- ❖ how Pre-mature mRNA become stable and mature molecules (**Post-transcriptional modifications including 5' capping, splicing, and 3' polyadenylation**)
- ❖ steric hindrance is **Physical obstruction caused by the spatial arrangement of atoms or groups within a molecule.**
- ❖ Later in **1990, Nancy Sturm and Larry Simpson** found that minicircles also encode gRNAs. But besides the **coding potential**, they found direct evidence for the existence of gRNAs.
- ❖ The **trp operon** consists of **a promoter, an operator, a leader (trpL), an attenuator, and five structural genes** designated as trpE, trpD, trpC, trpB and trpA.
- ❖ Although mRNA's short lifetime may seem wasteful, it has **an important regulatory function.**
- ❖ CRISPRs consist of repeat sequence...**30bp.. long**
- ❖ The Variable Loop: It sits between the anticodon loop and the ψ U loop and, as its name implies, **varies in size from 3 to 21 bases.**
- ❖ Eukaryotic cells always use **5'-AUG-3'** as the start codon.
- ❖ Regulatory domain of most activators interact with the _ **one or more of these complexes and recruit them to the gene.**
- ❖ type of DNA polymerase has 3'-5' exonuclease activity _____ **ExoI/ DNA polymerase I**
- ❖ component of the spliceosome machinery is recruited by the SR protein _____ **recruiting them to the nearby splice sites.**
- ❖ BER can repair the damage of DNA caused by.... Process _____ **deamination, oxidation, and alkylation**
- ❖ Enzymes are not involved in RNA editing _____ **DNA polymerase**

- ❖ Which is not a type of RNA (gRNA)
- ❖ Base pairing in DNA (A-T, C-G)
- ❖ Repeating units of nucleic acid (nucleotides)
- ❖ How many classes of histone (5)
- ❖ RNA editing uses enzymes except DNA polymerase
- ❖ are non coding genes of DNA rRNA, tRNA, etc/ Introns
- ❖ point out the wrong statement about alkylation damage to DNA. A) nitrogen and oxygen atoms external to the base ring systems; B) non-bridging oxygen atoms in phosphate groups; and C) nitrogen atoms in the base ring systems except those linked to deoxyribose. Chose other than these.
- ❖ the trp operon consists of which parts. promoter, operator
- ❖ contains small circular DNA with associated proteins. Plasmid
- ❖ steric hinderance is? can bind wala option hoga
- ❖ which poly merase enzyme show 3'» 5' exonuclease activity? DNA polymerase
- ❖ the hydrophilic backbone of sugar and phosphate face outside in DNA. false
- ❖ “Inchworming” invokes a flexible element within the polymerase that allows a module at the front of the enzyme, containing the active site.
- ❖ Pre mature mRNA becomes mature by which process. Spliceosome
- ❖ The site is the sote where peptide bond is formed. P site
- ❖ Who proposed the operon model. Manod and Jacob proposed
- ❖ Linus Pauling and Robert Corey carefully analyzed the peptide bond.
- ❖ Non coding sequence introns
- ❖ PROMOTER _ Contain legible element in polymerase which show module in front of enzyle
- ❖ Plasmid _ contain small circular DNA with associated protein
- ❖ Who proposed the operon model Manod and Jacob
- ❖ symmetry involving rotation about a single axis cyclic symmetry
- ❖ Which medium in which cell grow Culture medium
- ❖ Mismatch repair system ?/ Cells with a nonfunctional mismatch repair system have a high rate of mutation due to their inability to efficiently repair base pair mismatches
- ❖ Two of these, enzyme I (E-I) and the histidine containing protein (Hpr), are also components of other sugar transporter systems and therefore are unlikely to be direct participants in a glucose-specific phenomenon in Catabolite repression.
- ❖ m in mRNA stands for: messenger
- ❖ Griffith observed the phenomenon of Transformation in the bacterium Streptococcus pneumonia.
- ❖ Mus musculus mice was used by Griffith.
- ❖ Hydrogen bonds are formed between adjacent segments of polypeptide chain.
- ❖ Which kind of mutations don't cause more drastic changes in DNA: transversions.
- ❖ Water-mediated deamination converts guanine to xanthine.
- ❖ Production of a mono-adduct in which a single methyl group attaches to DNA.
- ❖ Two mammalian 3' exonucleases, TREX1 and TREX2 may correct the errors introduced by Pol β.
- ❖ A promoter is the DNA sequence that initially binds to the RNA polymerase.
- ❖ Inchworming invokes a flexible element within the polymerase that allows a module at the front of the enzyme, containing the active site.
- ❖ Rho-dependent terminators have rather ill-defined RNA elements called rut sites.
- ❖ In vitro, the general transcription factors are all that are required, together with Pol II, to initiate transcription on a DNA template.
- ❖ Transcriptional regulatory proteins called activators help recruit polymerase to the promoter, stabilizing its binding there.
- ❖ New RNA is recognized by an RNase called, in yeast, Rat1.
- ❖ True statement: *The most highly conserved sequences are the GU in the 5' splice site, the AG in the 3' splice site, and the A at the branch site.*
- ❖ 5' splice site recognized by the U1 & U6 snRNA.
- ❖ Steric Hindrance If the splice sites within the intron are too close together, splicing factors cannot bind to both sites at the same time.

- ❖ **Larry Simpson and colleagues** discovered guide RNAs (**gRNAs**).
- ❖ Eukaryotic cells always use **5'-AUG-3'** as the start codon.
- ❖ It is the site of attachment of the amino acid on tRNA. **The Acceptor Stem.**
- ❖ 23. The accumulation of toxic proteins, which result from misfolding may lead to several neurological disorders including ***Alzheimer's, Huntington's and Parkinson's diseases.***
- ❖ CRISPRs are repeated sequences, each **30 bp** long and highly conserved within a given cluster.
- ❖ mRNA's short lifetime may seem wasteful, it has an important **regulatory function.**
- ❖ **Monod and Jacob** proposed the operon model in 1961.
- ❖ The trp operon consists of a promoter, an operator, a leader (trpL), an attenuator, and **five structural genes designated as trpE, trpD, trpC, trpB and trpA.**
- ❖ The site on DNA where a repressor binds is called an **operator.**
- ❖ 29. The addition of acetyl groups to histone tails to **"loosen" chromatin structure, freeing up sites.**
- ❖ In a signal transduction pathway, the initiating ligand is typically detected by a specific cell surface receptor: the ligand binds to an extracellular domain of the receptor, and this binding is communicated to the intracellular domain. From there, the signal is relayed to the relevant transcriptional regulator, often through a cascade of **kinases.**
- ❖ One clear advantage of the lambda phages over plasmid vectors is that **they can accommodate much more foreign DNA.**
- ❖ Which one of these amino acids are present in high percentage in histone protein. **(Lycine)**
- ❖ During the removal of introns phosphodiester bonds are broken, and two new ones are made. **(No energy required)**
- ❖ During the splicing reaction..... **(Large amount of energy required)**
- ❖ The spliceosome machine size is similar to a? **(Ribosomes)**
- ❖ The ____ uptake channel allows ribonucleotides to enter the active center. **(Non Template)NTP channel**
- ❖ Within the active center cleft, the DNA strands separate from position **(+3)**
- ❖ Alternative splicing strategy enables a gene to give rise to more than one polypeptide product. These alternative products are called _____. **(Isoforms)**
- ❖ It is estimated that 90% or more of the protein-coding genes in the human genome are spliced in alternative ways to generate more than one isoform. **(90% or more)**
- ❖ the defined sequences are centered respectively at _____ and _____ upstream of the site where RNA synthesis starts **(10 bp and 35 bp)**
- ❖ Which factor recognize that from which point the transcription have to start **(Sigma)**
- ❖ Bacterial Core enzyme is consist of **(Alpha 2 Beta Beta Prime Omega)**
- ❖ In the case of E. coli the predominant Sigma factor is called? **(sigma 70)**
- ❖ **Larry Simpson and colleagues** found that editing system should add or delete UMPS when they discovered. **(gRNAs)**
- ❖ Information in nucleotides interpreted to generate sequence of amino acids is called. **(Translation)**
- ❖ The mammalian muscle protein is? **(Troponin T)**
- ❖ The CTD contains a series of repeats of the _____ sequence. **(heptapeptide)**
- ❖ Series of repeats of the heptapeptide sequence in yeast Pol II CTD are **(27)**
- ❖ Series of repeats of the heptapeptide sequence in worm Pol II CTD are? **(32)**
- ❖ Series of repeats of the hepta-peptide sequence in human Pol II CTD are? **(952)**
- ❖ Series of repeats of the hepta-peptide sequence the number of repeats correlate with the complexity of the _____. **(Genome)**
- ❖ Hepta-peptide sequence repeat contains sites for phosphorylation by specific enzyme called. **(Kinase)**
- ❖ The 3 step of Transcription is; **(Termination)**
- ❖ in some cells specific, well characterized sequences trigger termination in other it's is _____ what instructs the enzyme. **(Less clear)**
- ❖ Regulating the _____ state of the CTD of Pol II controls subsequent steps elongation and the processing of RNA. **(Phosphorylation)**
- ❖ Enzyme the remove phosphate group from CTD is called. **(Phosphatase)**

- ❖ TAF42 and TAF62 from *Drosophila* have been shown to form a structure similar to that of the ____ tetramer of Histones. (**H3.H4**)
- ❖ Structural studies suggest that segments of TFIIB insert into the RNA-exit channel and active center cleft of Pol II in a manner analogous to the α region ____ linker in the bacterial case. ($\frac{3}{4}$)
- ❖ Work of Franklin & Wilkins (**1950s**)
- ❖ Hershey-Chase experiments (**1952**)
- ❖ Avery's Experiments (**1944**)
- ❖ Griffith's Experiments (**1928**)
- ❖ They consist of two sequence elements:- A short inverted repeat of **20 nucleotides** followed by a stretch of about **eight A:T** base pairs.
- ❖ Transcriptional regulatory proteins called **activators** help recruit polymerase to the promoter, stabilizing its binding there.
- ❖ It works by binding to sites near the 3' splice site of specific exons in those pre mRNAs and recruiting the **U2AF splicing factor**.
- ❖ One important clue about the **mechanism of RNA editing** is that partially edited transcripts have been isolated, and these are always edited at their **3'-ends but not at their 5'-ends**.
- ❖ This suggests strongly that editing proceeds in a 3'→5' direction. Kenneth Stuart and colleagues first reported this phenomenon in **1988**.
- ❖ Then, in **1994**, Scott Seiwert and Stuart used a mitochondrial extract and a gRNA to edit a synthetic premRNA.
- ❖ The protein-coding region of the mRNA consists of an ordered series of three nucleotide-long units called **codons** that specify the order of amino acids.
- ❖ This element is also referred to as a **Shine Óalgarno** sequence after the scientists who discovered it by comparing the sequences of multiple mRNAs.
- ❖ The **Acceptor Stem**: It is so-named because it is the site of attachment of the amino acid, is formed by pairing between the **5' and 3'** ends of the tRNA molecule.
- ❖
- ❖
- ❖ Once the polymerase has transcribed the length of the gene (or genes), it must stop and release the RNA product (as well as dissociating from the DNA itself). This step is called **termination**.
- ❖ It is the addition of an initiation factor called σ that converts core enzyme (**$\alpha 2 \beta \beta' \omega$**) into the form that initiates only at **promoters**. This form of the enzyme is called the **RNA polymerase holoenzyme**.
- ❖ In the case of *Escherichia coli*, the predominant σ factor is called **$\sigma 70$** .
- ❖ Promoters recognized by polymerase have **two** conserved sequences each of **6 nucleotides**, separated by a nonspecific stretch of **17–19** nucleotides.
- ❖ The two defined sequences are centered, respectively, **at ~10 bp** and **at ~35 bp** upstream of the site where RNA synthesis starts. The sequences are thus called the **minus 35** and **minus 10** regions or element.
- ❖ The $\sigma 70$ factor can be divided into **four regions** called σ region 1 through σ region 4.
- ❖ Two helices within region 4 form a common DNA-binding motif called a **helixturn-helix**.
- ❖ In case of the bacterial enzyme bearing $\sigma 70$, this transition is often called **isomerization**.
- ❖ RNA polymerase performs two proofreading functions on that growing transcript. The first of these is called **pyrophosphorolytic editing**.
- ❖ second proofreading mechanism, called **hydrolytic editing**
- ❖ The sequences called **terminators** trigger the elongating polymerase to dissociate from the DNA and release the RNA chain it has made.
- ❖ The first, requires a protein called **Rho** to induce **termination**.
- ❖ Rho-dependent terminators have rather ill-defined RNA elements called **rut sites**.
- ❖ Rho-independent terminators are also called **intrinsic terminators**.
- ❖ When polymerase transcribes an inverted repeat sequence, the resulting RNA can form a stem-loop structure often called a "**hairpin**" by base-pairing with itself.
- ❖ a series of other elements called **silencers**

- ❖ The complete set of general transcription factors and polymerase, bound together at the promoter and poised for initiation, is called the **pre-initiation complex**.
- ❖ Many Pol II promoters contain a so-called **TATA** element (some 30 bp upstream of the transcription start site).
- ❖ The TATA element is recognized by the general transcription factor called **TFIID** (“TFII” denotes a transcription factor for Pol II).
- ❖ The component of TFIID that binds to the TATA DNA sequence is called **TBP**
- ❖ The other subunits in this complex are called **TAFs**
- ❖ Regions of **TFIIB** (called the **linker** and **reader**) aid in open complex formation, perhaps by **stabilizing** the melted DNA until the RNA:DNA hybrid takes over that role.
- ❖ Transcriptional regulatory proteins called **activators** help recruit polymerase to the promoter, stabilizing its binding there.
- ❖ P-TEFb phosphorylates and thereby activates another protein, called **SPT5**, itself an elongation factor.
- ❖ Finally, **TAT-SF1**, yet another elongation factor, is recruited by P-TEFb.
- ❖ Another class of elongation factor is the so-called **ELL family**.
- ❖ an enzyme that degrades the second RNA as it emerges from the polymerase has been identified, and this enzyme may itself trigger termination. This is called **the torpedo model of termination**.
- ❖ This new RNA is recognized by an RNase called, in yeast, **Rat1** (in humans, **Xrn2**) that is loaded onto the end of the RNA by another protein (**Rtt103**) that binds the CTD of RNA polymerase.
- ❖ Although the torpedo model for termination is now the favored one, there is an alternative called the **allosteric model**.
- ❖ initiation requires two other factors, called **SL1 and UBF**.
- ❖ Some Pol III promoters (e.g., those for the tRNA genes) consist of two regions, called **Box A and Box B**, separated by a **short element**.
- ❖ The coding sequences are called **exons** and the intervening sequences are called **introns**.
- ❖ The process of intron removal is called **RNA Splicing**.
- ❖ **Alternative splicing** strategy enables a gene to give rise to more than one polypeptide product. These alternative products are called **isoforms**.
- ❖ The exon intron boundary that is, the boundary at the 5’ end of the intron — is marked by a sequence called the **5’ splice site**.
- ❖ The intron exon boundary at the 3’ end of the intron is marked by the **3’ splice site**.
- ❖ Third sequence necessary for splicing. This is called the **branchpoint site** (or branchpoint sequence).
- ❖ The transesterification reactions are mediated by a huge molecular “machine” called the **Spliceosome**.
- ❖ This complex comprises about **150 proteins** and **five RNAs** and is similar in size to a **ribosome**.
- ❖ The five RNAs (U1, U2, U4, U5, and U6) are collectively called **small nuclear RNAs (snRNAs)**. Each of these RNAs is between **100 and 300** nucleotides long in most **eukaryotes** and is complexed with several proteins.
- ❖ These RNA – protein complexes are called **small nuclear ribonuclear proteins (snRNPs)** - pronounced “snurps”).
- ❖ BBP also called **SF1**.
- ❖ The former subunit interacts with BBP (SF1) and helps that protein bind to the branch site. This arrangement of proteins and RNA is called the **early (E) complex**.
- ❖ U2 snRNP then binds to the branch site, aided by U2AF and displacing BBP (SF1). This arrangement is called the **A complex**.
- ❖ Together, these three snRNPs U4,U5,U6 are called the **tri-snRNP particle**.
- ❖ The rearrangement brings together within the spliceosome those components that together form the active site. **C complex**
- ❖ The so-called Group I and Group II are **selfsplicing introns**.
- ❖ So-called SR (**Serine Arginine Rich**) proteins bind to the sequences called **exonic splicing enhancers (ESEs)** within the exons.
- ❖ identical U5 component is found in both the major and the alternative or so called **minor spliceosome**.

- ❖ In genome-wide studies, the most commonly seen forms of alternative splicing are cases in which complete exons are included or excluded from the mature message. Such exons are often called **cassette exons**
- ❖ Proteins that regulate splicing bind to specific sites called **exonic (or intronic) splicing enhancers (ESE or ISE) or silencers (ESS or ISS)**.
- ❖ Each SR protein has another domain, rich in arginine and serine, called an **RS domain**.
- ❖ We have emphasized alternative splicing as a way in which multiple protein products can be produced from a single gene. These different proteins are called **isoforms**.
- ❖ This kind of RNA editing is directed by an enzyme **called adenosine deaminase acting on RNA (ADAR)**.
- ❖ In particular, a very malignant human brain tumor called **glioblastoma multiforme** (GBM) has very low ADAR2 activity
- ❖ Export takes place through a special structure in the nuclear membrane called the **nuclear pore complex**.
- ❖ These RNAs are called **transfer RNAs (tRNAs)** because their attached amino acid is subsequently transferred to the growing polypeptide chain.
- ❖ The protein-coding region of the mRNA consists of an ordered series of three nucleotide-long units called **codons** that specify the order of amino acids.
- ❖ Enzymes called **aminoacyl-tRNA synthetases** couple amino acids to specific tRNAs that recognize the appropriate codon(s).
- ❖ The protein-coding region(s) of each mRNA is composed of a contiguous, nonoverlapping string of codons called an **open reading frame (commonly known as an ORF)**.
- ❖ To facilitate binding by a ribosome, many prokaryotic ORFs contain a short sequence upstream (on the 5' side) of the start codon called the **ribosome-binding site (RBS)**.
- ❖ Eukaryotic mRNAs recruit ribosomes using a specific chemical modification called the **5' cap**, which is located at the **extreme 5' end of the mRNA**.
- ❖ Once bound to the mRNA, the ribosome moves in a **5' 3' direction** until it encounters a 5'- AUG-3' start codon, a process called **scanning**.
- ❖ The reaction to form a new peptide bond is called the **peptidyl transferase reaction**.
- ❖ The initiation of prokaryotic translation commences with the small subunit and is catalyzed by three translation initiation factors called **IF1, IF2, and IF3**.
- ❖ The completion of translocation requires the action of a second elongation factor **called EF-G**.
- ❖ stop codons are recognized by proteins called **release factors (RFs)** that activate the hydrolysis of the polypeptide from the peptidyl-tRNA.
- ❖ Prokaryotes have two class I release factors called **RF1 and RF2**.
- ❖ In eukaryotic cells, there is a single class I release factor called **eRF1** that recognizes all three stop codons.
- ❖ Prokaryotes and eukaryotes have only one class II factor called **RF3 and eRF3**, respectively.
- ❖ this three-amino-acid sequence is called a **peptide anticodon** and must interact with and recognize stop codons.
- ❖ stop codons are recognized by proteins called **release factors (RFs)** that activate the hydrolysis of the polypeptide from the peptidyl-tRNA.
- ❖ In bacteria, cotranslational protein folding is assisted by a **48 kDa** chaperone called the **trigger factor**, which binds at the bacterial ribosome's exit pore.
- ❖ One class of bacterial regulatory RNAs (called **sRNAs**)
- ❖ RybB is expressed from a promoter recognized by a special σ factor called **σ^E (like σ^S , a stress response σ factor)**.
- ❖ The most striking examples are the **so-called** that control metabolic operons and in biosynthetic operons.
- ❖ transcription is terminated before polymerase has a chance to transcribe the downstream protein-coding segment of the gene. This form of transcriptional regulation is also called **attenuation**
- ❖ At one end of the array is a so-called leader sequence, often **A-T rich** and **500 bp** in length.

- ❖ CRISPRs have been found in half of all bacterial genomes sequenced, and essentially all genomes of Archaea.
- ❖ Clustered Regularly Interspaced Short Palindromic Repeats (or CRISPRs).
- ❖ The sequence in the virus that will become a new spacer is called the proto-spacer and is found close to a PAM.
- ❖ Those made artificially or produced in vivo from dsRNA precursors are typically called small interfering RNAs (SiRNAs).
- ❖ This machinery includes a complex called the RNA-induced silencing complex (RISC).
- ❖ Argonaute is often called Slicer and mRNA cleavage is called slicing.
- ❖ This DNA region contains a conspicuous duplication of a 72-bp sequence, called the 72-bp repeat.
- ❖ A DNA segment corresponding to an ORF plus the translational start and stop signals for protein synthesis is called a cistron and the mRNA encoding a single polypeptide is called monocistronic mRNA.
- ❖ The untranslated RNA sequence before the coding region is called the 5' – untranslated region (5'- UTR) or 5' – leader while the untranslated sequence after the coding region is called 3'– untranslated region (3'- UTR).
- ❖ This phenomenon, which is called coordinate regulation, results from control of the synthesis of a single polycistronic mRNA that encodes all of the gene products.
- ❖ This complementation test showed that all of the mutants initially isolated fell into two groups which were called lacZ and lacY.
- ❖ Other enzymes, called repressible enzymes, exhibit a decreased rate of synthesis in response to the addition of a small molecule in the medium.
- ❖ Still other enzymes, called constitutive enzyme, are synthesized at fixed rates under all growth conditions.
- ❖ users without being substrates of β galactosidase. □ Inducers having this property are called gratuitous inducers.
- ❖ Monod & Jacob called the lacI-gene product, the Lac repressor.
- ❖ A lacI+ / lacI – partial diploid has one good copy of the lacI gene product, so the system is inhibited.
- ❖ A reaction of this type in which an antibody that is raised in response to one protein is used to detect a closely related protein is called a cross-reaction and closely related protein is called crossreacting material (CRM).
- ❖ The lacO site is called the operator.
- ❖ The operator is unoccupied and the promoter is available for initiation of mRNA synthesis. This state is called derepression.
- ❖ the isolated Lac repressor does not bind lactose but does bind a lactose isomer called allolactose.
- ❖ Transcription-level inhibition of the lactose enzymes and a variety of other inducible enzymes by glucose (or other readily used carbon source) is called catabolite repression.
- ❖ This protein, called the cAMP receptor protein (CRP) or the catabolite activator protein (CAP), is encoded by the crp gene.
- ❖ Mechanism by which glucose influences lac operon transcription is called inducer exclusion.
- ❖ Sankar Adhya and coworkers solved the problem by showing that a histone-like protein called HU binds to the gal promoter region only when the Gal repressors are bound at both galOE or galOI causing the DNA to loop.
- ❖ Promoters like P1 in gal operon that have AS at position – 41.5 are called class II cAMP-dependant promoters.
- ❖ Constitutive mutants exhibit a sixfold increase in tryptophan enzyme synthesis when bases 123 – 150 within the leader are deleted. This 28 base sequence is called the attenuator.
- ❖ Segment 2 is therefore free to interact with segment 3 to form 2•3 hairpin (the so called antiterminator).
- ❖ This gives a low level of constitutive expression called the basal level.
- ❖ The site on DNA where a repressor binds is called an operator.
- ❖ This mechanism, often called recruitment, is an example of cooperative binding of proteins to DNA.

- ❖ In bacteria, we saw that genes activated by recruitment (such as the lac genes) can be activated in so-called **activator bypass experiments**
- ❖ it creates specific binding sites on nucleosomes for proteins bearing **so-called bromodomains**.
- ❖ Gal4 appears to contact at least three components: Mediator, TFIID, and a third complex called **SAGA** (Spt-AdaGcn5- acetyltransferase)
- ❖ P-TEFb is brought to the stalled polymerase by an RNA-binding protein called **TAT**.
- ❖ Histone (and DNA) modifications also form the basis of a type of repression called **silencing**.
- ❖ The DNA segment to be cloned is first linked to a **vector DNA**, which is a vehicle for carrying foreign DNA into a suitable host cell, such as the bacterium E. coli. □ Such carriers are called **vectors**.
- ❖ the MCS is part of a gene (LacZ) in its own right and codes for a portion of a polypeptide called **β-galactosidase**.
- ❖ The functional enzyme is able to hydrolyse a colourless substance called **Xgal** into a blue insoluble material.
- ❖ Another vector designed especially for cloning large DNA fragments is called a **cosmid**.
- ❖ These are like the cosmids in that they have characteristics of both phages and plasmids; thus, they are called **phagemids**.
- ❖ Downstream promoter elements (known as **DPE, DCE, and MTE**).
- ❖ The minor form of the spliceosome is also known as the **AT-AC spliceosome**
- ❖ The genetic information contained within the order of nucleotides in messenger RNA (mRNA) is interpreted to generate the linear sequences of amino acids in proteins. This process is known as **translation**.
- ❖ The first and last codons of an ORF are known as the **start and stop** codons.
- ❖ mRNAs containing multiple ORFs are known as **polycistronic** RNAs and those encoding a single ORF are known as **monocistronic** RNAs.
- ❖ This phenomenon of linked translation between overlapping ORFs is known as **translational coupling**.
- ❖ Because most amino acids are specified by more than one codon, it is not uncommon for one synthetase to recognize and charge more than one tRNA (known as **isoaccepting tRNAs**).
- ❖ The ribosome is composed of two subassemblies of RNA and protein known as the **large and small subunits**.
- ❖ Translation initiation is the only time a tRNA binds to the P-site without previously occupying the A-site. This event requires a special tRNA known as the **initiator tRNA**.
- ❖ This is not the case, however, because an enzyme known as a **deformylase** removes the formyl group from the amino terminus during or after the synthesis of the polypeptide chain.
- ❖ Two auxiliary proteins known as **elongation factors** control these events.
- ❖ This domain is known as **the factor binding center**.
- ❖ The promoter has two critical regions, one of which is the core element, also known as the **initiator (rINR)**, is located at the start of transcription, between positions **- 45 and +20**.
- ❖ The other is the **upstream promoter element** (UPE), located between positions **-156 and - 107**.
- ❖ In the early years of the cloning era, Boyer and his colleagues developed a set of very popular vectors known as the **pBR plasmid series**.
- ❖ The recombinant phage is then assembled into preformed viral protein particles, a process known as in **vitro packaging**.
- ❖ YACs containing a megabase or more are known as “**megaYACs**.”
- ❖ During initial transcription, RNA polymerase produces and releases short RNA transcripts of **<10 NUCLEOTIDES**
- ❖ Promoters recognized by polymerase have two conserved sequences each of 6 nucleotides, separated by a nonspecific stretch of **17–19** nucleotides.
- ❖ Kenneth Stuart and colleagues first reported this phenomenon in **1988**.
- ❖ Larry Simpson and colleagues found the answer in **1990** when they discovered **guide RNAs (gRNAs)** encoded in **Leishmania maxicircles**.

- ❖ in **1994**, Scott Seiwert and Stuart used a mitochondrial extract and a gRNA to edit a synthetic premRNA
 - ❖ Max Birnstiel and colleagues in **1991** demonstrated that transcripts of a bacterial neomycin gene transplanted into monkey **COS1** cells remained in the nucleus.
 - ❖ So Francis H. Crick in **1955** proposed that before their incorporation into polypeptides.
 - ❖ Monod and Jacob proposed the **operon model** in **1961** to explain how the lac system is regulated.
 - ❖ Walter Gilbert and Benno Muller-Hill successfully isolated the Lac Repressor from E. coli extracts in **1966**.
 - ❖ The Lac repressor is a homotetramer with a molecular mass of 154 kDa.
 - ❖ One useful, though somewhat dated, class of plasmids is the **pUC series**.
 - ❖ Charon 4 can accept up to about **20 kb of DNA**.
-
- ❖ Bacteria have only a single RNA polymerase, whereas eukaryotic cells have _____ RNA Polymerase _____ 3
 - ❖ Which of the following have three RNA polymerases (RNA Pol I, II, and III). Eukaryotes
 - ❖ Small interfering RNAs in plants are transcribed by _____ DNA-dependent RNA polymerases
 - ❖ The first step is the initial binding of polymerase to a promoter to form what is called a _____ closed complex.
 - ❖ A _____ is the DNA sequence that initially binds to the RNA polymerase in initiation of transcription. _____ Promoter.
 - ❖ The process involved in the RNA formation on the DNA template is called _____ Transcription
 - ❖ Which of the following statements are TRUE? _____ The mRNA is translated from 5' to the 3' end.
 - ❖ Which statement is true for elongation? _____ Unwinding of DNA, Proof reading, Dissociation of growing RNA, All given.
 - ❖ The bacterial core RNA polymerase starts transcription at _____ region.... Promoter
 - ❖ The enzyme required for transcription is _____ RNA polymerase
 - ❖ σ factor in Escherichia coli, is called _____ $\sigma 70$
 - ❖ Promoter site on DNA _____ initiates transcription
 - ❖ transition from the closed to the open complex involves structural changes in the _____ enzyme
 - ❖ In case of the bacterial enzyme bearing $\sigma 70$, this transition is often called _____ isomerization
 - ❖ σ acts as a molecular mimic of _____ DNA
 - ❖ Region 1.1 of σ is highly _____ charged _____ **negatively charged**.
 - ❖ RNA polymerase can initiate a new RNA chain on a DNA template without _____ primer
 - ❖ The Elongating _____ Is a processive machine that synthesizes and proofreads _____. polymerase-RNA
 - ❖ Only _____ nucleotides of the growing RNA chain remain base-paired to the DNA template at any given time-----8—9
 - ❖ During DNA elongation, the enzyme polymerase adds one _____ at a time to the growing _____ **Nucleotide-RNA**
 - ❖ The termination of transcription is a normal and important function at the ends of **gene/s**.
 - ❖ The sequences called **terminators** trigger the elongating polymerase to dissociate from the DNA and release the RNA chain it has made.
 - ❖ an unusual termination may be triggered by damaged _____ DNA
 - ❖ The hairpin works as an efficient **terminator**
 - ❖ Rho-independent terminators are also **called intrinsic terminators** because they need no other factors to work.
 - ❖ The process of transcription is **identical** in both the prokaryotes and eukaryotes.
 - ❖ **Bacteria** have only one RNA polymerase but all **eukaryotes** have at least three different ones i.e., Pol I, II, and III;
 - ❖ Plants have following polymerase except pol0

- ❖ The component of **TFIID** that binds to the TATA DNA sequence is called TBP (TATA binding protein).
- ❖ Pol II promoters contain a so-called TATA element (some 30 bp **upstream** of the transcription start site).
- ❖ The TATA element is recognized by the general transcription factor called **TFIID**
- ❖ The large subunit of Pol II has a **carboxyterminal domain (CTD)**, which is referred to as the “tail”. The CTD contains a series of repeats of the heptapeptide sequence: Tyr-Ser-Pro-Thr-Ser-Pro-Ser.
- ❖ The number of heptapeptide repeats (Tyr-Ser-Pro-Thr-Ser-Pro-Ser) in CTD correlates with the complexity of _____ Genome
- ❖ Several of the TAFs have structural homology with **histone proteins**,
- ❖ TFIIB, This protein, a single polypeptide chain, enters the **preinitiation complex** after TBP
- ❖ TFIIS also contributes to proofreading by **polymerase**.
- ❖ TFIIF acts as an **ATP**-driven translocator of double-stranded DNA.
- ❖ Crystal structure of the head module of yeast Mediator reveals that it contains **seven** subunits
- ❖ High regulated levels of transcription in vivo, require transcriptional regulatory proteins, the **Mediator complex**,
- ❖ Transcriptional regulatory proteins called **activators** help recruit polymerase to the promoter, stabilizing its binding there.
- ❖ Once polymerase has escaped the promoter and initiated transcription, it shifts into the **elongation phase**.
- ❖ The CTD tail is also very long. It could potentially extend **800 Å** from the body of the enzyme.
- ❖ The first human ELL protein was originally identified as the product of a gene that undergoes translocations in **acute myeloid leukemia**.
- ❖ Another factor that does not affect initiation, but stimulates elongation, is **TFIIS**.
- ❖ _____ Elongation factor bind to elongating polymerase and suppress transient pausing by the enzyme..... ELL family.
- ❖ polyadenylation takes place at **3'** end of the mRNA
- ❖ The final RNA processing event is **polyadenylation**...
- ❖ According to this model, termination depends on a **conformational change** in the elongating polymerase.
- ❖ The promoter for the **rRNA** gene comprises **two parts**: the core element and the UCE (upstream control element).
- ❖ Which of the following types of DNA polymerase has 3'→5' exonuclease activity? **DNA polymerase I**
- ❖ Some Pol III promoters (e.g., those for the **tRNA genes**) consist of two regions, called Box A and Box B, separated by a short element.
- ❖ The introns and exons are distinguished through _____. 5' and 3' splice site
- ❖ Messenger RNAs containing different selections of exons can be generated from a given pre-mRNA through a process called **alternative splicing**.
- ❖ The intron – exon boundary at the **3'** end of the intron is marked by the 3' splice site
- ❖ During RNA editing, the branchpoint site is present on _____. Intron
- ❖ There are generally _____ splicing sequence on pre-mRNA, 3
- ❖ Introns are removed from primary transcripts by cleavage at specific sequences called **splice sites**.
- ❖ The first reaction triggered during the removal of intron is _____ **2'-OH of the conserved A at the branch site**
- ❖ The transesterification reactions are mediated by a huge molecular “machine” called the **Spliceosome**.
- ❖ Which of the following are the snRNPs in splicing?
- ❖ They recognize the 5' splice site and the branch site
- ❖ They bring splice sites together as required
- ❖ They catalyze the RNA cleavage and joining reactions
- ❖ Ans= 1, 2 and 3

- ❖ Self-splicing means that: **The intron itself folds into a specific conformation within the precursor RNA and catalyzes the chemistry of its own release.**
- ❖ In the splicing pathway, 5' splice site is recognized by: U1
- ❖ The intron uses an _____ residue within the branch site to attack the phosphodiester bond during self splicing _____ A
- ❖ The nuclear pre-mRNA and group II intron splicing share the same _____.
Transesterification reactions
- ❖ There are a total of _____ classes of splicing found in the cell __3
- ❖ Which statement regarding the self splicing intron is incorrect? _ self-splicing introns are enzymes.
- ❖ In group II intron splicing intron is removed in the form of lariat. TRUE
- ❖ The structure of group I introns includes a binding pocket that will accommodate any **guanine nucleotide** or nucleoside
- ❖ in self splicing, group I introns use free __G__ nucleotides for splicing
- ❖ Group I introns contain an internal guide sequence that base-pairs with _____. 5' splice site sequence.
- ❖ the 5' splice site must be recognized initially by the U1 snRNP and then by the **U6 snRNP**.
- ❖ Which of the following statement regarding the splice sites is incorrect? _____ SR proteins are not critical for splicing.
- ❖ the 5' splice site must be recognized initially by _____. U1 snRNP and then by the U6 snRNP
- ❖ The minor form of the spliceosome is also known as the **AT-AC spliceosome**
- ❖ Despite the different **splice** site and **branchsite** sequences recognized by the two systems, these major and minor forms of spliceosomes both remove introns using the same chemical pathway.
- ❖ Which of the following components of the alternative spliceosome have the same roles in the splicing reaction as U1 and U2
- ❖ U11 and U12
- ❖ Find the incorrect statement regarding alternate splicing from the following: _____
Mammals use the minor splicing machinery to direct splicing of the majority of their pre-mRNAs.